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Differential involvement of right and left hemisphere in individual recognition in the domestic chick

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Abstract

Right hemisphere advantage in individual recognition (as shown by differences between response to strangers and companions) is clear in the domestic chick. Chicks using the left eye (and so, thanks to the complete optic decussation, predominantly the right hemisphere) discriminate between stranger and companion. Chicks using the right eye discriminate less clearly or not at all. The ability of left eyed chicks to respond to differences between strangers and companions stimuli is associated with a more general ability to detect and respond to novelty: this difference between left and right eyed chicks also holds for stimuli which are not social partners. The right hemisphere also shows advantage in tasks with a spatial component (topographical learning; response to change in the spatial context of a stimulus) in the chick, as in humans. Similar specialisations of the two hemispheres are also revealed in tests which involve olfactory cues presented by social partners. The special properties of the left hemisphere are less well established in the chick. Evidence reviewed here suggests that it tends to respond to selected properties of a stimulus and to use them to assign it to a category; such assignment then allows an appropriate response. When exposed to an imprinting stimulus (visual or auditory) a chick begins by using right eye or ear (suggesting left hemisphere control), and then shifts to the left eye or ear (suggesting right hemisphere control), as exposure continues. The left hemisphere here is thus involved whilst behaviour is dominated by vigorous response to releasing stimuli presented by an object. Subsequent learning about the full detailed properties of the stimulus, which is crucial for individual recognition, may explain the shift to right hemisphere control after prolonged exposure to the social stimulus. There is a

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marked sex difference in choice tests: females tend to choose companions in tests where males choose strangers. It is possible that this difference is specifically caused by stronger motivation to sustain social contact in female chicks, for which there is extensive evidence. However, sex differences in response to change in familiar stimuli are also marked in tests which do not involve social partners. Finally, in both sexes there are two periods during development in which there age-dependent shifts in bias to use one or other hemisphere. These periods (days 3–5 and 8–11) coincide with two major changes in the social behaviour of chicks reared by a hen in a normal brood. It is argued that one function of these periods is to bring fully into play the hemisphere most appropriate to the type of response to, and learning about, social partners which is needed at particular points in development. Parallels are discussed between the involvement of lateralised processes in the recognition of social partners in chicks and humans.

Key words: Right hemisphere; Left hemisphere; Domestic fowl; Lateralization; Chick

Introduction

A major characteristic of the social organization of adult domestic fowl is the establishment and maintenance of dominance relationships. This requires individual recognition of flock members, a capacity that seems to be based largely on visual cues (see Candland, 1969; Guhl and Ortman, 1953; Schelderupp-Ebbe, 1922).

Young domestic chicks also can distinguish familiar from unfamiliar conspecifics, even after only brief social experience (Zajonc et al. 1975; Rajecki et al. 1976; Jones, 1984). Although differences in response to familiar and unfamiliar conspecifics can, perhaps, be accounted for in terms of habituation (Zayan, 1987; Zayan and Lamberty, 1989), there is clear evidence (Ryan, 1982) that domestic fowl can learn to discriminate between the pictures of various unknown conspecifics, and this in conditions not associated with any particular social behaviour.

The present paper is concerned with the differential role (and use) of the two cerebral hemispheres in social recognition, and with sex differences in both lateralization and type of response to social stimuli.

Young chicks need, from very early in life, to respond appropriately to releasers presented by the mother. Subsequently, they need to elaborate representations of their main types of social partner (initially: hen, sibs) which stresses the invariant features of each type, rather than features which vary between individuals in relative random ways. Secondly, since chicks are capable of individual recognition (see below), they must be able to elaborate representations based on the unique properties of individuals.

Here we present evidence that the right hemisphere predominates in chicks in the elaboration and use of detailed descriptions of individual social partners.

Lateralization of control of response to visual change in a familiar social partner

Birds offer considerable advantages in the study of visual lateralization. When only one eye is in use, direct visual input is confined to contralateral brain structures. Crossover at

the thalamic level, in the chick (Rogers and Sink, 1987), as in other birds, allows indirect ipsilateral input to forebrain neurons. However, independent scanning movements of the eyes are usual in birds (Wallman and Pettigrew, 1985); as a result even the frontal visual fields (which are used in binocular fixation when the eyes are converged) will commonly each provide an independent (and quite different) input to the contralateral forebrain. As a result, monocular vision produced experimentally by covering one eye is less abnormal in birds than in mammals. Differences between chicks using the right or left eye are likely to reflect differences between the contralateral structures fed by the two eyes. Since these differences need not be confined to the forebrain, but might for example also depend upon right/left differences in the optic tecta, the terms 'right and left eye systems' (RES and LES) will be used when discussing purely visual responses in the chicks. In work on humans, it is conventional to use the terms right or left hemisphere to stand for the structures (no doubt largely at least in the forebrain) which show lateral specialisation. Where comparison with human evidence is being made, the same usage will be followed here. This is also more convenient in a number of other cases (e.g. when considering lateralisation of response to olfactory stimuli). Chicks using the right or left eyes will be distinguished as RE or LE, and chicks using both as BIN ('binocular').

The domestic chick shows marked hemispheric specialisation in response to visual stimuli (Rogers and Anson, 1979; Andrew et al. 1982; Andrew, 1983, 1988, 1991; see also below). We report here work (Vallortigara and Andrew, 1991; Vallortigara, 1992) in which we examined the ability of RE, LE and BIN chicks to distinguish between familiar and strange social companions. In these and all other experiments discussed here, each chick was tested under only one eye condition (RE, LE or BIN). We began with chicks reared in pairs and tested (on day 3) binocularly or with only one eye in use in a companion/stranger choice apparatus. At testing, each chick was placed centrally in a runway. The cagemate ('companion') and an unfamiliar chick ('stranger') from the same batch were visible behind glass in small compartments at opposite ends of the runway. Times spent in each third of the runway (ends and middle) were recorded. In both sexes, BIN and LE showed clear and significant, but different choice: males chose the stranger and females the companion (see Section 5 for discussion). RE males and females showed no clear choice (Fig. 1).

Similar results were obtained (Vallortigara and Andrew, 1991), using male chicks reared singly with an artificial social partner, (a red table tennis ball, with a white horizontal bar on one face, suspended at about chick head height). At test on day 3, each chick had to choose between a ball identical with that in the home cage and a ball which was changed in some way. The two balls were placed at opposite ends of the runway. When the ball was changed by the rotation of the bar to the vertical, both BIN and LE chose the familiar ball. A lesser rotation through 45 degrees, to an oblique position, caused LE to choose the novel ball instead. RE, on the other hand, did not show preference under either condition. Similar experiments, with females, revealed a different pattern: whatever the transformation, BIN, LE and RE all chose the familiar ball. The absence of choice, in RE males, under all these conditions (and in RE females when faced with choice between a companion and a stranger chick) did not reflect a lack of motivation to approach a social companion. Time spent by LE and RE chicks in the middle third of the runway was similar under all experimental conditions (see Vallortigara and Andrew, 1991).

Males, when relying on the LES, thus choose to approach small transformations of the social partner but to avoid large ones. Comparable shifts from interest and exploration to avoidance and fear with increased novelty has long been recognized (Schneirla, 1985).

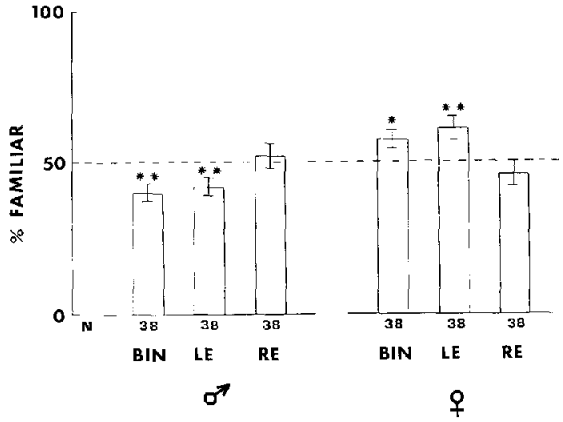

Fig. 1. Choice between familiar chick (cage companion) and unfamiliar chick. Scores are shown as (time near familiar/ total time near both familiar and unfamiliar)×100, with S.E.M. N indicates the number of chicks. Departures from chance (50%) are shown by asterisks (* $P < 0.05$, ** $P < 0.01$: one-sample two-tailed t-tests). An ANOVA showed females to choose the familiar chick more than did males (Sex: $F_{2,222} = 12.16$ $P < 0.001$). The pattern of choice differed with eye in use (BIN = binocular, LE = left-eyed, RE = right-eyed) in a different way in males and females (Sex x Eye: $F_{2,222} = 9.02$ $P < 0.010$). In both sexes, the absence of choice in RE differed significantly from the clear choice in BIN (Males: $F_{1,74} = 6.90$ $P = 0.010$; Females: $F_{1,74} = 5.23$ $P = 0.027$) and LE (Males: $F_{1,74} = 7.11$ $P = 0.009$; Females: $F_{1,74} = 8.07$ $P = 0.005$). (Redrawn from Vallortigara and Andrew, 1991.)

Choice of 'optimal discrepancy' has been also postulated to explain some features of mate selection (Bateson, 1979). Males, when relying on the RES, appear to ignore small or moderate transformations. When such birds do choose, they could be argued to have judged the transformed object not to be a social companion at all, since they then avoid it.

It is noteworthy that male chicks show a comparable pattern of lateralization when tested with small peckable objects (glass beads presented on the tip of a fine rod). When such an object was presented repeatedly in a standard schedule pecking curves showed very replicable habituation. The effects on such pecking curves of initial experience of an object, which differed in varying degree fron the standard object, gave a measure of the extent to which the two objects were judged to correspond by the chick (Andrew, 1983, 1991). LE males showed no transfer of habituation when the two objects differed in quite minor ways. Thus when the two objects differed only in the point in the cage from which they approached LE males treated them as if they were entirely different objects, whereas RE males showed almost complete transfer of habituation. LE were also responsive to changes in the characteristics of the bead itself (e.g. in colour) which RE ignored (Andrew, 1991), even though there is little doubt that these changes were readily detectible.

It is thus most economical to suppose that the LES (right hemisphere) of the chick is able to detect differences between familiar and strange social partners as part of a wider specialisation for the analysis and record of all the properties of a stimulus. Such wider specialisation is particularly clear for the analysis of spatial relations, where LE show markedly better performance than RE in tests requiring orientation by topographical cues (Rashid and Andrew, 1989). Very much the same can be said of the human right hemisphere, where specialisation for face recognition is accompanied by considerable advantage in the analysis of general spatial relations (Bradshaw and Nettleton, 1983).

Lateralization of response to olfactory change in a familiar social partner

Although lateralization in birds and mammals has been demonstrated in a number of visual and auditory tasks (review: Bradshaw and Rogers, 1993) different senses have not been compared in similar tasks. We have recently shown that in the male chick use of the right hemisphere allows detection of olfactory change in a familiar social companion, when use of the left hemisphere does not (Vallortigara and Andrew, 1994). Females were not tested.

Our procedure was similar to that used with visual stimuli. This time, however, olfactory transformations of the social companion were used. Chicks were reared singly from the first day of life in cages containing a red plastic cylinder (50 mm height × 33 mm), suspended by a fine thread at about chick head height. The cylinder had four 4 mm holes equally spaced along its equator, through which scent could escape from an internal cottonwool ball. In the first two experiments male chicks were reared with an odourless cylinder. At test, on day 3, they were placed centrally in a runway, at each end of which was hung a cylinder visually identical to that in the home cage. One cylinder contained cottonwool moistened with *n*-amyl-acetate (Experiment 1) or amyl-acetate or orange oil (Experiment 2). About 30 min before test, some chicks had one nostril temporarily closed by malleable wax. Note that, whereas the decussation of visual pathways means that in the chick, as in other vertebrates, visual inputs are to the contralateral hemisphere, primary olfactory inputs are not crossed.

In Experiment 1, chicks using their right nostril (RN) chose the familiar (scentless) object, whilst chicks using their left nostril (LN) associated indifferently with both objects (Fig. 2). The same was true in Experiment 2; here the additional group with both nostrils in use behaved like RN and chose the familiar object (Fig. 2).

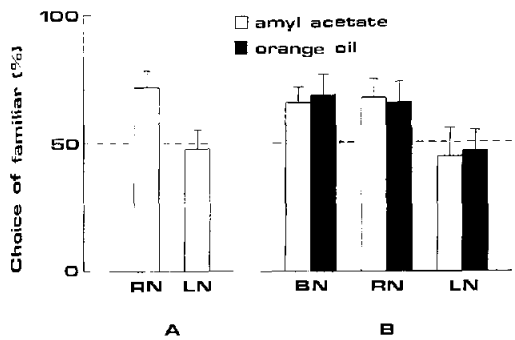

Fig. 2. Choice of the unscented (familiar) object (mean percentage of total time spent near both objects, with S.F.M.). Chicks used right (RN), left (LN) or both (BN) nostrils. In Experiment 1 (**A**) amyl acetate was the scent used at test, whilst in Experiment 2 (**B**) the scent was amyl acetate for some of the chicks and orange oil for others. In Experiment 1 chicks using their right nostril (RN) chose the familiar (scentless) object ($t_{26} = 3.53$ $P = 0.001$, one-sample two-tailed t-test), but those using their left nostril (LN) did not ($t_{27} = -0.27$), the difference between the two groups being significant (ANOVA, $F_{1,53} = 6.22$ $P = 0.015$). In Experiment 2 chicks using either both (BN) or the right nostril (RN) chose the scentless object ($t_{36} = 3.90$ $P < 0.001$; $t_{39} = 3.50$ $P = 0.001$, respectively). The ANOVA showed only the effect of nostril in use to be significant ($F_{2,109} = 5.04$ $P = 0.008$). LN chicks differed significantly from BN ($F_{1,73} = 7.40$ $P = 0.008$) and from RN ($F_{1,76} = 6.77$ $P = 0.010$). (Redrawn from Vallortigara and Andrew, 1994.)

In another experiment chicks were reared either with a cylinder scented with clove oil or with an unscented cylinder. Again LN failed to choose, whereas RN chose the cylinder which smelt like that with which they had been reared.

In a final experiment chicks were repeatedly presented with a small box of fine wire mesh at the end of a rod, at intervals of about 10 min. Within the box was a small plug of cottonwool scented with amyl acetate. Most RN chicks showed by their behaviour (head shaking or bill wiping) that they had detected a scent on the first trial, whereas LN chicks did not show such behaviour until the second or third trial. It is thus likely that olfactory lateralization, like visual, affects performance in tests which do not involve social companions.

Attachment to bedding odour has already been shown in the chick (Jones and Gentle, 1985). Taken with our evidence, this suggests that olfactory cues may play some part in early attachments, perhaps to the nest site, perhaps to the mother or sibs.

Effects of lateralization on social pecking evoked by familiar or unfamiliar conspecifics

The social pecking test (Rajecki et al. 1976; Zajonc et al. 1975) takes advantage of the marked xenophobia exhibited by young socially-reared chicks towards unfamiliar conspecifics, which become the targets of strong aggressive pecking. We used this procedure to investigate recognition of familiar and unfamiliar conspecifics in pair-reared chicks, tested either binocularly or with only one eye in use (Vallortigara, 1992). Males and females were tested on day 3, in tournaments of pairs composed of either cagemates or strangers. For each pair of companion chicks, one other pair of companion chicks was selected to form a tetrad. For each tetrad, the tournament involved observing each individual in several encounters (3 min each) against each of the remaining three, according to a specified order. The results (Fig. 3) parallel those obtained in the experiments which have already been discussed, in which chicks had to choose between an object like that in the home-cage and some transformation of it. Both LE and BIN groups treated strangers and companions differently; RE treated them similarly. However, unlike what was found in the choice tests, here both sexes behaved similarly. Both pecked the stranger more than the companion.

The total number of pecks, lumped across stranger and companion, was virtually identical in RE and LE of the same sex (mean $\pm$ S.E. were: males: LE = 73.0 $\pm$ 7.8; RE = 74.3 $\pm$ 12.3; females: LE = 46.9 $\pm$ 7.7; RE = 42.6 $\pm$ 5.5). This suggests a sex difference independent of patterns of lateralisation, with males tending to peck other chicks more than females do.

Sex differences in the response to social stimuli

When able to use both eyes, males and females show clear, but opposing, patterns of choice of social partner: males tend to spend more time near a stranger and females to spend more time near a companion (above). This unexpected finding seemed to explain why some studies (below), which did not take account of the sex of the subjects, had failed to demonstrate any ability of young chicks to discriminate between social partners.

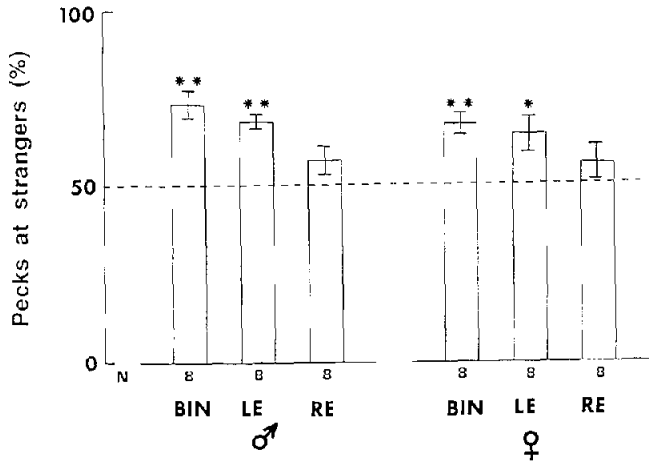

Fig. 3. Percentages of pecks at strangers (group means ± S.E.M. are shown). N indicates the number of tetrads. Significant departures from chance level (50%) are shown (** $P < 0.01$, * $P < 0.05$, one-sample two-tailed t-tests). The ANOVA showed a significant effect of the eye in use ($F(2,42) = 5.42$ $P < 0.01$). Paired comparisons revealed a significant difference between RES and LES ($F(1,30) = 4.67$ $P < 0.05$) and between BIN and RES ($F(1,30) = 10.51$ $P < 0.01$), but not between BIN and LES ($F(1,30) = 1.04$ n.s.). (Redrawn from Vallortigara, 1992.)

In general, we would expect clear evidence for discrimination of strangers, even when sex is neglected in analysis, in those cases where both sexes show the same sort of difference between response to a stranger and to a companion. Pecking at strangers is one such case. Other examples include food pecking, which is about five times higher in the presence of companions than of strangers, and levels of distress calling, which is markedly higher when among strangers than when among cagemates (Rajecki et al., 1977).

Even choice tests, which have not taken sex into account, have provided clear evidence of discrimination, when extreme differences between stranger and companion have been used: Salzen and Cornell (1969), for instance, reported that week-old chicks could learn to discriminate between familiar and unfamiliar colours of conspecifics when artificially dyed, but that they did not discriminate between familiar and unfamiliar undyed conspecifics of the same breed. The latter finding agrees with Sigman, Lovern and Schulman (1978) who reported no preference in socially reared 48 hours old chicks when allowed a simultaneous choice between either a familiar or an unfamiliar socially reared chick. The appearance of choice when dyed strangers are used is not unexpected on our hypothesis: with so large a change as this, males as well as females might be expected to choose the companion (Vallortigara and Andrew, 1991).

Somewhat similar discrepancies between studies are present in the literature on open-field tests, in which the tested chick is either accompanied by a stranger or by a companion. Again, differences between the effect of stranger and companion tend to be found only when sex is taken into account. Suarez and Gallup (1983) observed that chicks (sex not distinguished) froze and remained silent longer when in pairs than when single, but whether the pair member was a cagemate or a stranger seemed to be unimportant. In contrast, Jones (1984) found that 7-day-old female chicks, reared in pairs and tested in the open field in the presence of a cagemate, showed significantly shorter durations of freezing, and vocalized and walked significantly sooner than did chicks tested with strangers.

Cailotto et al. (1989) repeated Jones's procedure but used both males and females: they found that, whilst females indeed showed lower latencies when tested with cagemates than when tested with strangers, males showed no differences between the two conditions.

Open-field tests, although widely used in attempts to measure differences in the readiness with which fear responses are evoked in different groups of animals, are notoriously difficult to interpret. A major reason is that locomotion can result both from attempts to escape and from a return from fear-induced immobility to investigation and maintenance behaviour. A specific hypothesis has been provided for the chick by Gallup and Suarez (1980), who proposed that here open-field behaviour reflects a variable balance between opposing tendencies to reinstate social contact with conspecifics (locomotion and attempts at escape) and to evade predation (freezing). Vallortigara and Zanforlin (1988) used this model to predict the sex differences which should be obtained in the chick in open field tests. They argued that females should be more strongly affected by separation from social fellows, because of a stronger attachment to their fellows: females indeed began to walk and to give distress-calling sooner than did males. This seemed not to be due to a lesser readiness in females to show fear responses, since the sexes did not differ in their response to introduced predator stimuli. Further confirmation that the sex difference was associated with differences in the effect of social separation, rather than in response to frightening change in the environment was provided by manipulating exposure to social isolation prior to the open field test. As animals became accustomed to social isolation, so sex differences disappeared in the open field test.

There is other evidence that female chicks may try harder than males to avoid separation from social companions. Workman and Andrew (1989) found that in broods containing equal numbers of males and females and reared under semi-natural conditions, female chicks consistently remained closer to the mother and were less likely to go out of sight of her. Vallortigara et al. (1990) trained chicks to run along a runway to reach a goal box which contained either cagemates or food. Females ran faster to reach social partners, males to reach food. When the task required indirect approach, because of a sharp bend in the runway, the sex difference was reversed, apparently because the high motivation of the females to reach their fellows now interfered with successful completion of the task. Association with social companions remains strong in adult hens (Hughes, 1977; Dawkins, 1982; Bradshaw, 1991).

Many of the above findings have been duplicated within a single study, in which social discrimination was investigated using approach latencies, simultaneous free choice tests, and intersubject social pecking (Vallortigara, 1992). In approach latency tests, females showed shorter latencies when faced with cagemates than when faced with strangers, whereas males showed shorter latencies when faced with strangers than when faced with cagemates. In simultaneous-choice tests, females spent more time near cagemates, whereas males spent more time near strange chicks. This was true in both 3-day-old chicks, confirming the findings of Vallortigara and Andrew (1991), and in 7-day-old chicks.

Clearly field work on the later development and dispersal of broods is needed before the possible functions of these early sex differences can be established. They are consistent with a continuing close association between sisters, perhaps within their natal territory, and dispersal of individual males, attracted by strangers. It is also possible that the sex differences we have described are not important early in development, but instead represent an early appearance of features of adult behaviour. McBride and Foenander (1962) and McBride et al. (1969) have reported that, in natural populations of feral animals, single dominant cocks defend a large territory within which a number of females

live. Territorial males are bound to be very responsive to intruding strangers, whilst females have to be capable of strong bonding to their broods and perhaps also to the other hens with which they share the territory.

Sex differences in lateralization of response

Sex differences in the responses of chicks to companions and strangers can be explained in two ways: (1) a greater tendency for females to maintain contact with their social fellows; (2) a greater response of females to change in a familiar object, including social fellows. Alternative (2) could give rise to (1), in that, if the absence of a social fellow were assessed as a larger change by females, this in itself might be more disturbing. It is impossible to exclude the more complicated alternative, namely that both (1) and (2) coexist as independent effects. However, we examine here evidence for (2), from work with non-social stimuli, and consider whether (2) could explain our data for sex differences in response to social partners. We then discuss the possibility that the sexes differ in some feature of lateralisation.

To begin with the first issue: there are sex differences in patterns of response to coloured beads (Andrew and Dharmaretnam, in prep.), which resemble those which exist for social responses. The beads are too small to evoke social responses (Clifton and Andrew, 1983), so that the most economical explanation is that the basic sex difference is in lateralisation.

Briefly, the bead experiments (Andrew, 1991) established habituation curves for the evocation of pecking by a bead illuminated internally by violet light. The degree of transference of habituation from an earlier experience with a bead differing in various ways from the standard bead to subsequent tests with the standard bead was then measured. The smallest change in properties was a shift of wavelength of light from an initial red to the standard violet. Next, in degree of change came a change from unlit to violet illumination. The largest change was between a first experience in which the rod, on which the bead was normally mounted, was presented without the bead, so that at the presentation of the standard bead all bead properties were novel. Each chick was tested throughout as LE, RE or BIN.

In both sexes LE ceased to show transfer of habituation at smaller differences than did RE. In addition, LE males showed the same degree of habituation after prior presentation of a red bead as after a violet bead, as though illuminated beads presented in a standard way were treated as fully equivalent. Females did not transfer habituation in this way: red and violet beads were treated as largely different objects. There was a similar sex difference in the point in the series of transformations at which there was a complete disappearance of transfer of habituation: in females this occurred with change from the unlit bead to a violet-lit bead, whereas with males it occurred only when there was no bead (but only the rod tip) at the first presentation.

Findings of this sort, taken with the results of change in the social partner (above) are most economically explained by supposing that female chicks tend to assess any given degree of change as being greater than do males. If this were true, then one explanation of our failure to find choice of stranger in females would be that we did not present a small enough degree of change in the social companion for females to find the changed object interesting, rather than something to be avoided.

There was a second consistent sex difference (Andrew, 1991): when RE, LE and BIN birds were compared, in females the curves showing the course of habituation and the

degree of transfer of habituation shown by BIN were closely similar to those for either RE or LE. In males this was not true; instead BIN curves differed markedly from both monocular curves, usually being substantially higher than either RE or LE. This suggests that one of the eye systems is usually fully in control in binocular females when pecking a bead; the higher levels in binocular, than in monocular males may reflect involvement of both eye systems.

Stable control of one eye system by the other is also suggested by the viewing patterns evoked by different types of stimuli (including social partners, potential or actual: Dharmaretnam and Andrew, in press; Andrew et al. in preparation, a). Chicks tend to use one eye when viewing objects presented at a distance. Females show much clearer differences in choice of eye with which to view, according to the stimulus presented, than do males. Three types of stimuli were presented: a live hen, a live rat and a small light, all seen for the first time. The most extreme differences were between hen (RE) and light (LE) in females. In males there was no such extreme stimulus-specific use of either eye. It is clear from this study that there is marked right/left specialisation in females; indeed if this were the only type of study available one might argue that hemispheric specialisation was larger in females than in males. However, this is in direct contrast to the results of transfer of habituation tests (Andrew, 1991). Here, males typically show very clear diferences between RE and LE, in particular at the first presentation of the standard stimulus after prior experience with a stimulus differing in some way. A good example (Andrew, 1983) is the effect of change in the position in space from which the bead approaches. RE males show complete transfer of habituation despite such change, whereas LE males show almost no transfer, treating the two stimuli as quite different. In females, both RE and LE show no transfer of habituation at the first presentation of the standard stimulus (Andrew, unpublished). This is not because for some reason this is a transformation to which females are indifferent. The male type of difference appears at the next presentation, with LE females beginning to behave like LE males. This and other evidence to be presented elsewhere (Andrew et al. in preparation, b) show that sex differences in response to change are not confined to change in social partners.

There is not space here to consider the relationship between the various sex differences which have been considered here (see Andrew et al. in preparation, b). However, it is worth noting that it is possible that sex differences such as greater interest in maintaining contact with social companions are secondary consequences of differences in patterns of lateralization.

Changes in hemisphere involvement during first establishment of social attachments

We have recently found (Andrew et al. in preparation, a) that during first exposure to either a visual model partner (a table-tennis ball) or recorded hen clucks, chicks initially use the eye or ear of the right side of the head, and then, with further exposure, reverse to using the left side. The timecourse of change appears to be somewhat different for the two stimuli, with reversal taking several hours in the case of the cluck, but occurring sooner in the case of the table-tennis ball.

Greater involvement of the left hemisphere early in visual imprinting is also suggested by the fact that the structural change within forebrain structures, which is consequent on imprinting (i.e. consequent on first exposure to a potential visual social partner), is much more marked in the left hemisphere (review: Horn, 1991). However, it is unlikely that the

use of the right eye in our studies depends upon accompanying establishment of the first social attachment or on a relatively immature state of the CNS structures which are involved in imprinting. In chicks which lived with, and were attached to a partner chick, the same bias to right eye use was present at first encounter with a hen up to at least 15 days of age (Dharmaretnam and Andrew, in press).

We argue instead that left hemisphere predominance during first response to an object, which powerfully evokes response such as approach and social contact, is best understood in terms of the way in which the left hemisphere analyses perceptual input. Our hypothesis is that, at first exposure to an object like a hen, the chick is affected chiefly or entirely by those key releasing features which evoke approach and association, and that under these circumstances the left hemisphere tends to be in control. We have yet to test this idea further. It is consistent with the fact that the one situation in which RE chicks show marked advantage over LE is in the selection of food particles from amongst distracting background stimuli (Mench and Andrew, 1986; Zappia and Rogers, 1987). Here too the successful chicks are responding vigorously and rapidly to selected cues, rather than examining all the detailed properties of a particular complex stimulus. The later shift to right hemisphere involvement can be related much more certainly to the sort of right hemisphere specialisation which we have discussed in earlier sections. The only additional assumption that is needed is that learning about, and response to the subtle and varied features which allow individual social partners to be told apart, only become important after a period of exposure in which attention has been largely directed to conspicuous key features, in particular releasers.

Periods of bias to control by either right or left hemisphere during development and their possible relation to the recognition of social fellows

One further phenomenon in the chick, which suggests that lateralisation affects responses to social fellows, remains to be considered. This is the occurrence during behavioural development of two periods, with very standard timing (days 3–5 and 8–11), in each of which there are days of age-dependent bias to control by right or by left hemisphere (Dharmaretnam and Andrew, in press; Andrew et al., in preparation, b).

Age-dependent changes in hemispheric control were first suggested by the unusual sensitivity to pharmacological unilateral insult of the left hemisphere on day 8 of life, and of the right hemisphere on days 10 and 11 (Rogers and Ehrlich, 1983; Rogers, 1991). Subsequently, Workman and Andrew (1989) showed that, in broods kept under semi-natural conditions, changes in behaviour occurred at the times of the shifts in hemispheric dominance. These changes were appropriate to the specializations of the hemisphere which became dominant. For instance, chicks first begin to go out of sight of the mother hen and to explore the surroundings on day 10, when right hemisphere dominance would be expected to cause interest in topographical features (Rashid and Andrew, 1989).

In a recent study (Dharmaretnam and Andrew, in press), preferential eye use in viewing various stimuli at different ages was investigated. The overall timecourse of shifts in age-dependent bias was consistent with the timecourse, which has just been described (with RE use peaking on day 8 and LE use on day 11) and was very similar in males and females.

A variety of evidence (ease of distraction by laterally presented stimuli, choice of eye and ear to turn to stimuli associated with social companions: Andrew et al., in preparation,

b) suggest bias to left hemisphere control on days 3 and 5, and to right hemisphere control on day 4.

The two periods of shifting bias coincide with two major transitions in behaviour and experience of chicks, which are being reared in normal broods. Days 2–5 is the period in which the opportunity to see and the need to follow freely moving conspecifics first appear and then increase progressively. Day 10 is the first day of locomotion independently of, and away from the mother, which, as will be seen below, is preceded by special behaviour on day 8.

Overall, days of bias to left hemisphere control are associated with the points of first or greatly increased exposure to a new category (or a key feature of a category) of social companion. In section 7 it was noted that left hemisphere control predominates early in exposure to imprinting stimuli (with later shift to right hemisphere control). It thus seems that left hemisphere control is functionally important during initial experience of, and response to stimuli, which possess releasers strongly evoking social responses. Why this might be is considered later.

The evidence from the periods of age-dependent bias is as follows: The first such period begins with a marked increase in visual experience of moving siblings. Continuous video recordings of the behaviour of broods under a hen, showed that on day 1, all that happened was an occasional very brief protrusion of the head by one chick at a time. On day 2, visual experience of the hen became common as a result of increasing numbers of full emergences (about a 10-fold increase from day 1). Since emergence was often (at least in half of the cases) from under the hen's breast, and return was virtually always at the breast, chicks were likely to see the mother's head moving (and at the same time to hear it emitting clucks). At the same time there was some opportunity to see the largely motionless body of the mother. There was also for most chicks some opportunity to see one of their sibs: about two-thirds of emergences involved more than one chick.

The main new feature of day 3 was a substantial increase in the proportion of emergences which involved all the brood. On the average each chick emerged as part of the entire brood on about 12% of occasions on day 2, and on 46% of occasions on day 3. For information about days 4 and 5 it is necessary to turn to Workman and Andrew (1989), who report changes in locomotion and related behaviour in natural broods observed daily in a semi-natural habitat. On day 4 hens remained fully broody, although they moved short distances away from the nest. Chicks thus, for the first time, had some restricted opportunity to follow the mother, and probably even more experience of their sibs in movement than on day 3. However, much the sharpest change occurred between days 4 and 5. This was the time of the greatest single decrease in broodiness. For the first time hens moved freely, and chicks had to follow them repeatedly during the day.

Day 5 is thus the day when following, learning about and attaching to the moving mother may be of particular importance; the same may be true for moving siblings on day 3. Note that what is being argued is that there may be several steps in the establishment of visual social attachments during normal development, not that there is no visual attachment in the first two days of life; admittedly the scope of what could be learned and the time available to do it is likely to be very limited on day 1.

In the second period of bias-shift, brief strong bias to left hemisphere control on day 8 coincides with the equally brief appearance of staring at adult conspecifics other than the mother and at the human observer (Workman and Andrew, 1989). If, as seems possible, such viewing allows the chick to learn about unfamiliar conspecifics and other objects which move independently in the environment, this would be yet another example of an

association between left hemisphere control and first experience of some aspect of a category of social companion: here, the chick is being exposed to objects which need to be distinguished from its own mother and siblings.

Only experiments can decide what properties of the left hemisphere may explain the above associations. Two types of explanation deserve consideration. Firstly, left hemisphere control may help to sustain attention and response during early exposure to objects which not only present highly effective releasers, but are also presumably novel in almost every regard. Attempts to analyse all the characters of such a stimulus might preclude adaptive response. The second type of explanation is complementary: perhaps in very early experience it is best to learn by focussing on key features (here, probably releasers) rather than by storing descriptions of all properties simultaneously from the start.

The appearance of age-dependent bias to right hemisphere control on days 10 and 11 is almost certainly connected with the appearance on day 10 of movement in the environment which is independent of the mother. Andrew (1988) suggested that the interest in spatial context of stimuli and in topographical features, which is shown by chicks in which the right hemisphere is in control, might be one cause of the independent movement. This could be coupled with intensive learning about topography, which would be likely to be adaptive at the onset of relative independence. However, comparison with human evidence (Conclusions) suggests that right hemisphere control could also play a part in coping with the problems posed by meeting large numbers of strange conspecifics for the first time. One consequence of moving independently is an increased likelihood of meeting and interacting with chicks of other broods, and with their mothers. It may, in consequence, be important to pay special attention to individual differences between conspecifics at this time of development. Right hemisphere ability to identify individuals suggests that right hemisphere control would promote this.

Age-dependent-bias to right hemisphere control on day 4 is chiefly of interest at present as providing a second example of a shift in control from left to right hemisphere during development. Possible functions remain entirely speculative: perhaps the shift promotes learning of detailed and complete descriptions of mother (as seen sitting on the nest) and siblings.

In summary, there is evidence that, early in exposure to an imprinting stimulus, factors intrinsic to such an experience tend to bring the left hemisphere into play. We have argued that there are, in addition, age-dependent changes in bias, which promote such control at times when proper response and learning about potential social companions is crucial. At certain other times it is important that the right hemisphere should control and appropriate age-dependent bias promotes this also.

Conclusions

The left and right cerebral hemispheres of the chick appear to play very different roles in the recognition of, and responses to, social partners. It is likely that this is due primarily to different modes of analysis of perceptual input, rather than to hemispheric differences in the ease with which social behaviour is evoked. This is suggested by (1) the existence of similar hemispheric differences in tests using stimuli which do not evoke social responses; (2) the similar and high level of social contact shown by chicks using the right or left eye, despite the differences in pattern of choice which are shown in the same tests.

Our conclusions as to the nature of hemispheric involvement in social responses may be

summarised as follows: the left hemisphere is certainly capable of strong response to social partners; however, it is not competent to distinguish between individuals. One reason for this may be that left hemisphere analysis tends to build up a representation of categories of 'social partner', using chiefly those invariant features that belong to all members of each category. At least some of these features are likely to be selected as important because they are releasers of social behaviour, such as features of the head of the hen (see Johnson and Morton, 1991 for evidence in chick and in other species, including human infant).

The right hemisphere is particularly suited to build up representations based on past records of particular experiences which are as complete as possible. The assessment of novelty is clearly central to the identification of a stranger. As it is likely to be difficult or impossible to chose in advance the features which may be different in any particular stranger, it is necessary to have available full descriptions of familiar fellows, if strangers are to be distinguished from companions. These in turn must be are compared as fully as possible during analysis with the new object, if novelty is to be assessed. Our evidence suggests that olfactory and auditory properties, as well as visual, are used in this way by the right hemisphere.

The existence of periods of bias to control by one or other hemisphere at the two key times of major change in the behaviour and environment of the chick during normal rearing by a hen in a brood, is an indication of the probable importance of ensuring the involvement of the appropriate hemisphere at such times.

The lateralisation of function described here for the chick has obvious parallels in mammals. Humans (Bradshaw and Nettleton, 1983), and rhesus monkeys (Hamilton and Vermeire, 1988) also show a right hemisphere advantage in distinguishing familiar conspecifics. The chick may well, like humans (Johnson and Morton, 1991), rely heavily on facial features in recognition of natural conspecifics (Candland, 1969; Johnson and Horn, 1987). In humans this is paralleled by right hemisphere advantage in recognising other relatively unfamiliar patterns (De Renzi, 1982); this wider ability seems likely to be related to the specific right hemisphere advantage in facial recognition.

Periods of bias to control by one hemisphere may also occur in human development. Ladavas (1982) found that greater expressiveness on the left half of the face (under right hemisphere control) appeared in children at 10–11 years. Mann et al. (1979) showed that ability to recognise both voices and faces peak at 10 years, shows a trough in the period 12–13 or 14 years, and subsequently rises again. Levine (1985) specifically associated the peak in recognition with the right hemisphere by showing that left visual field advantage for unfamiliar faces appears in the period 8–10 years. The possibility of a period of overall bias to right hemisphere control around 10–11 years deserves examination; note that it differs from the commonly expressed idea that the hemispheres may differ in their rates of maturation, with special abilities being enhanced or acquired as maturation proceeds. Instead it would assume that basic specialisations are largely there at birth and that at least some age-dependent shifts in performance reflect shifts in which hemisphere is most likely to take charge. There is in fact evidence (de Schoner and Deruelle, 1991) for very early hemispheric specialisation: left visual field advantage in infants (4–9 months) for facial recognition, distinguishing a recently familiar face from that of a stranger, and for left visual field advantage in recognition of patterns using global rather than local features have all been reported. Molfese (1977) found that evoked potentials were greater in the left hemisphere for speech but not music in neonates.

If the changes at 10–11 years in humans are indeed comparable to those associated with the periods of right hemisphere bias in chick development, then a period (or periods)

of right hemisphere bias may be associated with preparations for beginning to leave the small groups within which early rearing occurs and begin to interact with (and so have to recognise as individuals) much larger numbers of conspecifics. It remains to be determined if there are increases in the ability to recognise individuals in chicks at the peak of bias to right hemisphere control (days 10 and 11).

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